Measure theory in a nutshell

- Not all measures have a density function → Densities exist only relative to another measure
- A measure μ is absolutely continuous w.r.t. another ν, written $\mu \ll \nu$ if $\nu(A) = 0 \Rightarrow \mu(A) = 0$

Radon–Nikodym Theorem:

If $\mu \ll \nu$ then there exists a function f such that: $\mu(A) = \int_A f(x)\, d\nu(x)$

This function $f = \dfrac{d\mu}{d\nu}$ is the Radon–Nikodym derivative

- When ν is the Lebesgue measure, $\mu \ll \nu$ means μ has a density $\rho(x)$
- This formalizes what we often write as: $d\mu(x) = \rho(x)\, dx$

Radon Measures